

MUST lecture 11

**Multiple-populations classification based on training data
with known population membership**

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Multiple-populations classification based on training data

Problem **Classify new multivariate observation** as being from **one of g populations** based on training data with known population membership

Training data

- already collected “ground truth” data with known/defined population membership

New data

- “real” new observations for which definitive establishment of population membership would be
 - too expensive
 - too slow
 - impossible
 - only possible by destructive test

Applications

- Pattern recognition / detection
- Machine learning
- Digital communication / detection of signals
- Medicine / pathology
- Feature extraction

Multiple populations classification: methodology (g populations)

Model

- $g \geq 2$ populations: $\pi_1, \pi_2, \dots, \pi_g$
 - π_i : $f_i(\mathbf{x}), \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, i = 1, \dots, g$
 - MVN assumption: $\mathbf{X}|\pi_i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), i = 1, \dots, g$
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Sample space

- p-dimensional multivariate observations, $\mathbf{x} = [x_1, \dots, x_p]^T \in R^p$
 - training data available for each population $\rightarrow \hat{f}_i(\mathbf{x}), \hat{\boldsymbol{\mu}}_i, \hat{\boldsymbol{\Sigma}}_i, i = 1, \dots, g$
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Classification

- splitting of sample space $R^p = \bigcup_{i=1}^g R_i, R_i \cap R_k = \emptyset, i \neq k$
- decision rule for new observations $d: \mathbf{x}_0 \in R^p \rightarrow d \in \{d_1, d_2, \dots, d_g\}$
where $d(\mathbf{x}_0) = d_i$ for $\mathbf{x}_0 \in R_i$

Multiple populations classification: informations used for decision making (g populations)

Misclassifications: Confusion matrix

Classify as →		d_1	d_2	...	d_g
True	π_1	$P(1 1)$	$P(2 1)$	...	$P(g 1)$
	↓ π_2	$P(1 2)$	$P(2 2)$	...	$P(g 2)$
	...	...	...	...	...
	π_g	$P(1 g)$	$P(2 g)$	...	$P(g g)$

where $P(k|i) = \int_{R_k} f_i(\mathbf{x}) d\mathbf{x}, \quad \mathbf{x} \in R^p$

Misclassifications: Cost matrix

Classify as →		d_1	d_2	...	d_g
True	π_1	0	$c(2 1)$	...	$c(g 1)$
	↓ π_2	$c(1 2)$	0	...	$c(g 2)$
	...	...	...	...	...
	π_g	$c(1 g)$	$c(2 g)$	...	0

where $c(k|i) \geq 0, \quad i \neq k$

Priors

- $p_i = P(\pi_i), \quad i = 1, \dots, g$
where $\sum_{i=1}^g p_i = 1$

General classification criteria (distribution-free / non-parametric)

ECM, Expected Cost of Misclassifications

$$ECM_{total} = \sum_{i=1}^g p_i \sum_{k=1}^g P(k|i)C(k|i), \quad k \neq i \quad (\text{total cost})$$

$$ECM(k) = \sum_{i=1}^g p_i P(k|i)C(k|i), \quad i \neq k \quad (\text{cost for classify as population } k)$$

Thus, choose decision k that minimizes $ECM(k)$ or equivalently:
Given new observation x_0 choose decision k that minimizes

$$ECM(k|x_0) = \sum_{i=1}^g p_i f_i(x_0)C(k|i), \quad i \neq k$$

Special case Equal costs: Choose k that maximizes $p_k f_k(x_0)$

Special case Equal costs and priors: Choose k that maximizes $f_k(x_0)$ (ML)

MAP criteria, Maximum A posteriori Probability (usefull for soft decisions)

Choose k that maximizes $P(\pi_k|x_0) = \frac{p_k f_k(x_0)}{P(x_0)} \propto p_k f_k(x_0)$

Classification for g Multivariate Normal (MVN) distributions

Model $X|\pi_i \sim N_p(\mu_i, \Sigma_i), \quad i = 1, 2, \dots, g$

Case 1: $\exists i, k: \Sigma_i \neq \Sigma_k$
(equal costs case)

Choose k that maximizes $p_k f_k(x_0)$

Put in MVN pdf and simplify giving

QDA, Linear Discriminant Analysis decision rule

Choose k that maximizes $\left\{ -\frac{1}{2} D^2(x_0, \mu_k) - \frac{1}{2} \log |\Sigma_k| + \log p_k \right\}$

Sample based (training data):

Put in estimates $\hat{\mu}_k = \bar{X}_k, \quad \hat{\Sigma}_k = S_k$

Case 2: $\forall \Sigma_i = \Sigma$
(equal costs case)

Choose k that maximizes $p_k f_k(x_0)$

Put in MVN pdf and simplify giving

LDA, Linear Discriminant Analysis decision rule

Choose k that maximizes $\left\{ \mu_k^T \Sigma^{-1} x_0 - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log p_k \right\}$

Sample based (training data):

Put in estimates $\hat{\mu}_k = \bar{X}_k, \quad \hat{\Sigma} = S_P$

Classification performance evaluation

APER, Apparant Error Rate

Make confusion matrix with counts of correct and incorrect classification of training data

$$APER = \frac{\# \text{ of misclassifications}}{\text{total \# of classifications}}$$

APER is **biased** and underestimates the true error rate because same data (training data) are used first to make decision rule and the again to assess classification succes

Estimated Actual Error Rate

Use cross-validation or hold-out procedure for one training data observation at a time

Make confusion matrix with counts of correct and incorrect classification

$$\widehat{AER} = \frac{\# \text{ of misclassifications}}{\text{total \# of classifications}}$$

The estimated Actual Error Rate is an **unbiased** estimate of the actual error rate

Summarizing several-population discriminant analysis procedures

Training data

- collect “ground truth” data
- define or estimate prior probabilities, $p_i = P(\pi_i)$, $i = 1, \dots, g$
- define misclassification costs, $c(k|i) \geq 0$, $i, k = 1, \dots, g$, $i \neq k$
- estimate parameters for $f(\mathbf{x}|\pi_i)$, i.e. $\hat{\boldsymbol{\mu}}_i, \hat{\boldsymbol{\Sigma}}_i$, $i = 1, \dots, g$
- MVN model control
- test $H_0: \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \dots = \boldsymbol{\Sigma}_g \ggg$ LDA case for accept / QDA case for reject
- test $H_0: \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 = \dots = \boldsymbol{\mu}_g \ggg$ significant separation of populations for reject
- specify/calculate sample based discriminant functions for LDA or QDA
- calculate confusion matrix for training data using the discriminant function
- calculate estimated classification error rates, APER and/or AER_{hat}

New data

- observe new data $\mathbf{x}_0$
- classify $\mathbf{x}_0$ as member of one of the populations $1, 2, \dots, g$ using the calculated discriminant functions for the training data
- calculate posterior probabilities using
 - $P(\pi_i|\mathbf{x}_0) \propto p_i f(\mathbf{x}_0|\pi_i)$, $i = 1, \dots, g$ (posterior_i $\propto$ prior_i·likelihood_i)
where $\sum_{i=1}^g P(\pi_i|\mathbf{x}_0) = 1$ (normalization to actual probabilities)